

STEP 1 — Install and Import Required Libraries

```
# Install required libraries (run once in terminal)
!pip install chromadb sentence-transformers

# Import libraries
import chromadb # Vector database
from sentence_transformers import SentenceTransformer # For creating embeddings
```

Collecting chromadb
 Downloading chromadb-1.5.5-cp39-abi3-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (7.2 kB)
 Requirement already satisfied: sentence-transformers in /usr/local/lib/python3.12/dist-packages (5.3.0)
 Collecting build>=1.0.3 (from chromadb)
 Downloading build-1.4.2-py3-none-any.whl.metadata (5.8 kB)
 Requirement already satisfied: pydantic>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.12.3)
 Requirement already satisfied: pydantic-settings>=2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.13.1)
 Collecting pybase64>=1.4.1 (from chromadb)
 Downloading pybase64-1.4.3-cp312-cp312-manylinux1_x86_64.manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_5_x86_64
 Requirement already satisfied: uvicorn>=0.18.3 in /usr/local/lib/python3.12/dist-packages (from uvicorn[standard]>=0.18.3-
 Requirement already satisfied: numpy>=1.22.5 in /usr/local/lib/python3.12/dist-packages (from chromadb) (2.0.2)
 Requirement already satisfied: typing-extensions>=4.5.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.15.0)
 Collecting onnxruntime>=1.14.1 (from chromadb)
 Downloading onnxruntime-1.24.4-cp312-cp312-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (5.2 kB)
 Requirement already satisfied: opentelemetry-api>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)
 Collecting opentelemetry-exporter-otlp-proto-grpc>=1.2.0 (from chromadb)
 Downloading opentelemetry_exporter_otlp_proto_grpc-1.40.0-py3-none-any.whl.metadata (2.6 kB)
 Requirement already satisfied: opentelemetry-sdk>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.38.0)
 Requirement already satisfied: tokenizers>=0.13.2 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.22.2)
 Collecting pypika>=0.48.9 (from chromadb)
 Downloading pypika-0.51.1-py2.py3-none-any.whl.metadata (51 kB)
 Requirement already satisfied: tqdm>=4.65.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.67.3)
 Requirement already satisfied: overrides>=7.3.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (7.7.0)
 Requirement already satisfied: importlib-resources in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.5.2)
 Requirement already satisfied: grpcio>=1.58.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (1.78.0)
 Collecting bcrypt>=4.0.1 (from chromadb)
 Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
 Requirement already satisfied: typer>=0.9.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.24.1)
 Collecting kubernetes>=28.1.0 (from chromadb)
 Downloading kubernetes-35.0.0-py2.py3-none-any.whl.metadata (1.7 kB)
 Requirement already satisfied: tenacity>=8.2.3 in /usr/local/lib/python3.12/dist-packages (from chromadb) (9.1.4)
 Requirement already satisfied: pyyaml>=6.0.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (6.0.3)
 Requirement already satisfied: mmh3>=4.0.1 in /usr/local/lib/python3.12/dist-packages (from chromadb) (5.2.1)
 Requirement already satisfied: orjson>=3.9.12 in /usr/local/lib/python3.12/dist-packages (from chromadb) (3.11.7)
 Requirement already satisfied: httpx>=0.27.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (0.28.1)
 Requirement already satisfied: rich>=10.11.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (13.9.4)
 Requirement already satisfied: jsonschema>=4.19.0 in /usr/local/lib/python3.12/dist-packages (from chromadb) (4.26.0)
 Requirement already satisfied: transformers<6.0.0,>=4.41.0 in /usr/local/lib/python3.12/dist-packages (from sentence-trans
 Requirement already satisfied: huggingface-hub>=0.20.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transform
 Requirement already satisfied: torch>=1.11.0 in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (2.10
 Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.6.1
 Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages (from sentence-transformers) (1.16.3)
 Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.12/dist-packages (from build>=1.0.3->chromadb) (2
 Collecting pyproject_hooks (from build>=1.0.3->chromadb)
 Downloading pyproject_hooks-1.2.0-py3-none-any.whl.metadata (1.3 kB)
 Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (4.12.1)
 Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (2026.2.2
 Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (1.
 Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx>=0.27.0->chromadb) (3.11)
 Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx>=0.27.0->ch
 Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
 Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20.0->
 Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.20
 Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->chromadb
 Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.12/dist-packages (from jsons
 Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.12/dist-packages (from jsonschema>=4.19.0->ch

STEP 2 — Initialize ChromaDB

```
# Create a client (starts the database)
client = chromadb.Client()

# Create a collection (like a table in DB)
collection = client.create_collection(name="rag_collection")
```

STEP 3 — Load Embedding Model

```
# Load pre-trained embedding model
model = SentenceTransformer('all-MiniLM-L6-v2')
```

```

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and fast
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to
modules.json: 100% 349/349 [00:00<00:00, 19.7kB/s]

config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 10.1kB/s]

README.md: 10.5k/? [00:00<00:00, 815kB/s]

sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.80kB/s]

config.json: 100% 612/612 [00:00<00:00, 38.1kB/s]

model.safetensors: 100% 90.9M/90.9M [00:01<00:00, 208MB/s]

Loading weights: 100% 103/103 [00:00<00:00, 463.28it/s, Materializing param=pooler.dense.weight]
BertModel LOAD REPORT from: sentence-transformers/all-MiniLM-L6-v2
Key | Status | |
-----+-----+--
embeddings.position_ids | UNEXPECTED | |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
tokenizer_config.json: 100% 350/350 [00:00<00:00, 14.1kB/s]

vocab.txt: 232k/? [00:00<00:00, 7.43MB/s]

tokenizer.json: 466k/? [00:00<00:00, 19.5MB/s]

special_tokens_map.json: 100% 112/112 [00:00<00:00, 9.32kB/s]

config.json: 100% 190/190 [00:00<00:00, 13.3kB/s]

```

STEP 4 — Prepare Sample Documents

```

# Sample data (documents)
documents = [
    "Artificial Intelligence is the simulation of human intelligence.",
    "Machine Learning is a subset of AI.",
    "Deep Learning uses neural networks.",
    "Natural Language Processing deals with text data.",
    "RAG combines retrieval and generation."
]

# Unique IDs for each document
ids = ["doc1", "doc2", "doc3", "doc4", "doc5"]

```

STEP 5 — Convert Text to Embeddings

```

# Convert all documents into vectors (embeddings)
embeddings = model.encode(documents)

```

STEP 6 — Store Embeddings in ChromaDB

```

# Store documents + embeddings in database
collection.add(
    documents=documents,
    embeddings=embeddings,
    ids=ids
)

```

STEP 7 — Perform Similarity Search

```

# Query input
query = "What is AI?"

# Convert query into embedding
query_embedding = model.encode([query])

# Search top 2 similar documents
results = collection.query(
    query_embeddings=query_embedding,

```

```
        n_results=2
    )

    # Print raw results
    print(results)

{'ids': [['doc1', 'doc2']], 'embeddings': None, 'documents': [['Artificial Intelligence is the simulation of human intelligence']]}
```

STEP 8 — Display Retrieved Results

```
# Print only the matched documents
for doc in results['documents'][0]:
    print(doc)
```

```
Artificial Intelligence is the simulation of human intelligence.
Machine Learning is a subset of AI.
```

STEP 9 — Build Simple Retrieval Function

```
# Function to retrieve relevant documents
def retrieve(query):
    # Convert query to embedding
    query_embedding = model.encode([query])

    # Search database
    results = collection.query(
        query_embeddings=query_embedding,
        n_results=2
    )

    # Return top results
    return results['documents'][0]

# Test the function
print(retrieve("Explain machine learning"))

['Machine Learning is a subset of AI.', 'Deep Learning uses neural networks.']
```